Breaking this into two parts, we see that the first sum telescopes to give $\mathrm{W}(\tau)^2$ (as $\mathrm{W}(0) = 0$). On the other hand, from (6.7), we have

$$\lim_{\nu\to\infty}\mathbb{E}\sum_{k=0}^{\nu-1}\Big[(\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t))^2\Big] = \tau,$$

but an even stronger statement is possible. Since we are summing a large collection of random variables $\Delta\mathrm{W}_k$ whose means are δt, the Central Limit Theorem tells us that, if we assume it to be well defined, the variance of the sum of squares from τ must tend to zero as a function of ν. Thus we may understand the convergence in a somewhat stronger sense than convergence in expectation. We make this precise in the following proposition

Proposition 6.2 *Let $\nu\delta t = \tau$ then*

$$\text{l.i.m.}_{\nu\to\infty}\sum_{k=0}^{\nu-1}\Big[(\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t))^2\Big] = \tau,$$

where, for a sequence of random variables, A_ν *and a random variable* B*, we have*

$$\text{l.i.m.}_{\nu\to\infty}\mathrm{A}_\nu = \mathrm{B} \Leftrightarrow \lim_{\nu\to\infty}\mathbb{E}(\mathrm{A}_\nu - \mathrm{B})^2 = 0.$$

(This is referred to as convergence in the mean square sense.)

Proof With τ, ν fixed, we have $\delta t = \tau/\nu$.

Let $\Delta\mathrm{W}_{\nu,k} = \mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t) = \mathrm{W}((k+1)\tau/\nu) - \mathrm{W}(k\tau/\nu)$ and observe that

$$\sum_{k=0}^{\nu-1}\Delta\mathrm{W}_{\nu,k}^2 - \tau = \sum_{k=0}^{\nu-1}\Big[\Delta\mathrm{W}_{\nu,k}^2 - \frac{\tau}{\nu}\Big].$$

Then

$$I_\nu \overset{\text{def}}{=} \mathbb{E}\left(\sum_{k=0}^{\nu-1}\Delta\mathrm{W}_{\nu,k}^2 - \tau\right)^2 = \mathbb{E}\left(\sum_{k=0}^{\nu-1}\Big[\Delta\mathrm{W}_{\nu,k}^2 - \frac{\tau}{\nu}\Big]\right)^2.$$

Terms of the form $Z_{\nu,k} \overset{\text{def}}{=} \Delta\mathrm{W}_{\nu,k}^2 - \frac{\tau}{\nu}$ are independent and have mean zero. Therefore

$$\mathbb{E}\left(\sum_{k=0}^{\nu-1}Z_{\nu,k}\right)^2 = \mathbb{E}\sum_{k=0}^{\nu-1}Z_{\nu,k}^2 - 2\mathbb{E}\sum_{k\neq l}Z_{\nu,k}Z_{\nu,l} = \mathbb{E}\sum_{k=0}^{\nu-1}Z_{\nu,k}^2,$$